

COP 4610 Operating Systems Principles

Project #1 (Shared Memory)

Sep 22, 2026

Project Objectives:

The purpose of this project is to introduce students to the concept of shared memory and the problems that can occur if shared memory is not protected adequately.

Total points Available: 100

Due: Oct 4 at 11:59 pm

Demo time: Oct 13

Project Description

In this assignment, you will create **4 processes**. Each of these processes will share a variable called "total". Each will increment the variable "total" by one to 100,000, 200,000, 300,000 and 500,000 respectively. Make sure that only the newly created child calls the function "process#()"

After **all the children have finished**, the parent process should release the shared memory and terminate. Use the **"wait"** function so that the parent knows precisely when each of the children finishes. The parent should print the process id of each child as the child finishes execution. Then it should release shared memory and print "End of Program".

You need to run the program several times and analyze your observations (write a detailed report).

Sample output

From Process 1: counter = 270547.
From Process 2: counter = 347860.
From Process 3: counter = 400001.
From Process 4: counter = 500000.

Child with ID: 2412 has just exited.
Child with ID: 2411 has just exited.
Child with ID: 2413 has just exited.
Child with ID: 2415 has just exited.

End of Simulation.

Submitting your assignment

- Submission via Canvas Assignment.
 - It is your responsibility to submit these assignments in a timely fashion.
- All files should be zipped together.
- There should be a readme file explaining in detail the exact steps to be taken to compile and execute the code files and the title page.
- Testing of this work should be done only on your VM for COP 4610.
- In case of any code errors, partial credit may be offered based on the code and documentation.
- A PDF report that presents the performance evaluation of your solution, and the complete LaTeX source folder (the submitted LaTeX source must compile successfully and reproduce the submitted PDF).
 - The report should be properly formatted (an academic format style, such as ACM or IEEE) and contain quantitative data along with your analysis of these data.
 - The final report must be written in LaTeX using either the provided IEEE or ACM template. Reports prepared using Word, or other formats will not be accepted.
 - If you do not submit your LaTeX source files or if your main .tex file cannot be compiled to generate the submitted PDF, your report will receive a grade of 0.
- AI Usage Documentation (15%):
 - a. If you choose not to use AI tools, you must still document how you could have used them (see below b).
 - b. If you use AI tools, you must submit a PDF report documenting your usage. This should include:
 - Prompts and Suggestions (e.g., provide the specific prompts you used)
 - Rationale for AI Usage:
 - Explain why you used particular prompts and how the AI-generated suggestions contributed to the project.
 - Describe how AI helped optimize or debug your implementation.
 - Incremental Development:
 - Show how you used AI tools to gradually improve your program, from initial implementation to refinement.
 - If you modified AI-generated code, explain why and how.
- Code Demo (10%):
 - You are required to demonstrate your program **in person** to the instructor or TA during the scheduled demo time, either during class or office hours. During the demo, you should be able to run your code, explain how it works, and answer basic questions about your implementation. You may also be asked to make a small modification to your code during the demo.
 - Each Code Demo will take approximately 2–3 minutes, with a maximum of 3 minutes.
 - To receive full credit for the Code Demo, your submitted code must compile, run successfully, and satisfy the assignment requirements.
 - The purpose of the Code Demo is to verify your understanding of the work you submitted. If you miss the deadline for completing the Code Demo, you will receive a zero for the Code Demo portion of the assignment unless prior approval has been granted or a documented exception applies.
 - Failure to adequately explain submitted code may result in loss of Code Demo points and, when appropriate, further review under the course Academic Integrity Policy.

Late Submission Policy

- For each day an assignment is late - up to a maximum of 2 days, the grade is reduced by 15%. For example, if you submit 90%-correct assignment 2 days late, your overall assignment score will be 60%.

Grading Criteria

- Minus 90% if code does not compile. Minus 70% if it compiles but does not run.
- If the code compiles and runs, further deductions will be made for the following:
 - Minus 40% if 4 children are not created.
 - Minus 40% if the processes are not cooperating processes.
 - Minus 40% if the children fail to modify the shared variable.
 - Minus 40% if the report is not written.
 - Minus 25% if the report contains no data analysis.
 - Minus 20% if parent ends without waiting for all children to exit.
 - Minus 10% if parent does not release shared memory before ending.
 - Minus 10% if children do not print out their results.
 - Minus 10% if parent does not print each time a child finishes.
 - Minus 10% if no comments.
 - Minus 5% if your name and username are not included in comments at the top of your source code

Development Environment

You may write your program using any available editor Nano, Emacs, Vi, or whatever editor you are most comfortable with, BUT, it must compile with gcc and be executable on your VM for COP 4610 (server: ocelot.aul.fiu.edu).

Tip

Build your project in an incremental fashion. Attempt to meet each objective before moving on to the next.